

Implementation and Analysis of an Optimal Rubik's Cube Solver using Kociemba's Two-Phase Algorithm with Real-Time Computer Vision

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Abstract

The Rubik's Cube, invented by Ernő Rubik in 1974, presents a formidable computational challenge with approximately 4.3×10^{19} possible configurations. Finding an optimal solution—one that reaches the solved state in the minimum number of moves—requires navigating an astronomically large state space. This paper presents the implementation and analysis of a real-time Rubik's Cube solver that combines Herbert Kociemba's Two-Phase Algorithm with computer vision capabilities for automatic cube state detection.

Our implementation achieves near-optimal solutions averaging 18-21 moves, computed within 500 milliseconds for typical random configurations. The system employs Iterative Deepening A* (IDA*) search with precomputed pruning tables to efficiently navigate the solution space. The solver supports multiple input modalities: webcam-based color detection using HSV color space analysis, manual keyboard entry, and interactive 3D visualization. Additionally, the system provides augmented reality overlays displaying step-by-step solving instructions directly on the video feed.

The Two-Phase Algorithm reduces the problem by first transforming any cube state into an intermediate subgroup G_1 , then solving to the identity state using restricted moves. This approach balances computational efficiency with solution quality, consistently producing solutions close to the theoretical minimum of 20 moves (God's Number) while maintaining practical response times suitable for real-time applications.

Keywords: Rubik's Cube, Two-Phase Algorithm, Kociemba, Group Theory, IDA*, Computer Vision, Augmented Reality

1 Introduction

1.1 Background

The Rubik's Cube, invented by Hungarian architect Ernő Rubik in 1974, has transcended its origins as a mechanical puzzle to become a subject of significant mathematical and

computational interest. The cube consists of 26 visible cubies (8 corners, 12 edges, and 6 fixed centers) that can be manipulated through rotations of the six faces: Up (U), Down (D), Left (L), Right (R), Front (F), and Back (B).

The total number of reachable configurations is given by:

$$N = \frac{8! \times 3^7 \times 12! \times 2^{11}}{12} = 43,252,003,274,489,856,000 \approx 4.3 \times 10^{19} \quad (1)$$

This astronomical figure arises from the permutations of corners ($8!$), corner orientations (3^7 , as the last corner's orientation is determined by the others), edge permutations ($12!$), edge orientations (2^{11}), and parity constraints (division by 12).

1.2 The Problem: God's Number

A fundamental question in Rubik's Cube research is: *What is the minimum number of moves required to solve any configuration?* This value, known as "God's Number," was proven to be exactly 20 in 2010 by Rokicki et al. [2]. Their proof required approximately 35 CPU-years of computation, demonstrating that while every cube position can be solved in 20 moves or fewer, finding such optimal solutions for arbitrary configurations remains computationally intensive.

The challenge lies in the trade-off between optimality and computational time:

- **Optimal solutions** require exhaustive search techniques that may take hours for complex configurations.
- **Practical solutions** should be computed in seconds while minimizing move count.

1.3 Motivation: Why Kociemba's Algorithm?

Several algorithmic approaches exist for solving the Rubik's Cube:

1. **Layer-by-Layer (LBL):** The human method, producing solutions averaging 100+ moves.
2. **Thistlethwaite's Algorithm:** Uses 4 phases to progressively restrict moves, averaging 45 moves.
3. **Kociemba's Two-Phase Algorithm:** Reduces to 2 phases, achieving near-optimal solutions (18-25 moves) in sub-second time.
4. **Optimal Solvers:** Guarantee 20 moves but require minutes to hours of computation.

Kociemba's algorithm provides the optimal balance for practical applications, delivering solutions within 1-5 moves of the theoretical optimum while maintaining real-time performance. This makes it ideal for applications requiring immediate feedback, such as our computer vision-based solver.

2 Literature Review

2.1 Layer-by-Layer Method

The most common human solving method, Layer-by-Layer (LBL), solves the cube in stages: first the cross, then corners of the first layer, middle layer edges, and finally the last layer using memorized algorithms. While intuitive, this method typically requires 80-120 moves and does not scale to computational optimization.

2.2 Thistlethwaite's Algorithm

Morwen Thistlethwaite introduced a breakthrough approach in 1981 [3] by decomposing the solution into four phases, each restricting the available moves:

$$G_0 = \langle U, D, L, R, F, B \rangle \rightarrow G_1 = \langle U, D, L, R, F^2, B^2 \rangle \quad (2)$$

$$G_1 \rightarrow G_2 = \langle U, D, L^2, R^2, F^2, B^2 \rangle \quad (3)$$

$$G_2 \rightarrow G_3 = \langle U^2, D^2, L^2, R^2, F^2, B^2 \rangle \quad (4)$$

$$G_3 \rightarrow G_4 = \{I\} \text{ (Identity/Solved)} \quad (5)$$

This approach guarantees solutions within 52 moves using lookup tables of manageable size.

2.3 Kociemba's Two-Phase Algorithm

Herbert Kociemba refined Thistlethwaite's approach in 1992 [1], reducing the four phases to two:

$$G_0 = \langle U, D, L, R, F, B \rangle \quad (6)$$

$$G_1 = \langle U, D, L^2, R^2, F^2, B^2 \rangle \quad (7)$$

$$I = \text{Identity (Solved State)} \quad (8)$$

By combining phases and using IDA* search with pruning tables, Kociemba's algorithm achieves solutions averaging 21 moves in under one second. The algorithm has become the de facto standard for practical cube-solving applications.

2.4 Optimal Solvers

Rokicki's work [2] proved God's Number using a combination of:

- Coset solving to reduce the search space
- Symmetry and antisymmetry exploitation
- Massive distributed computation (35 CPU-years)

While academically important, optimal solvers are impractical for real-time applications.

3 Mathematical Foundation: Group Theory

3.1 The Rubik's Cube Group

The set of all possible Rubik's Cube configurations forms a **group** under the operation of move composition. Let G denote this group:

$$G = \langle U, D, L, R, F, B \rangle \quad (9)$$

where each generator represents a 90° clockwise rotation of the corresponding face. This group has order $|G| \approx 4.3 \times 10^{19}$ and exhibits rich algebraic structure.

3.2 Two-Phase Decomposition

Kociemba's insight was to identify a subgroup $H \subset G$ such that:

$$H = G_1 = \langle U, D, L^2, R^2, F^2, B^2 \rangle \quad (10)$$

The subgroup H has order $|H| = 19,508,428,800 \approx 2 \times 10^{10}$, which is significantly smaller than G .

3.2.1 Phase 1: $G_0 \rightarrow G_1$

In Phase 1, we search for a sequence of moves that transforms the initial state into *any* element of the coset space G/H . Mathematically, Phase 1 solves:

$$\text{Find } m_1 \in G^* \text{ such that } s \cdot m_1 \in H \quad (11)$$

where s is the initial state and G^* is the free monoid generated by all moves.

The conditions for belonging to H are:

1. All edge orientations are zero (edges are "good")
2. All corner orientations are zero (corners are "good")
3. The four middle-layer edges (FR, FL, BR, BL) are in the middle layer

The coordinate representation captures these conditions:

$$\text{twist} \in \{0, 1, \dots, 2186\} \quad (3^7 = 2187 \text{ corner orientations}) \quad (12)$$

$$\text{flip} \in \{0, 1, \dots, 2047\} \quad (2^{11} = 2048 \text{ edge orientations}) \quad (13)$$

$$\text{slice} \in \{0, 1, \dots, 494\} \quad \left(\binom{12}{4} = 495 \text{ middle edge positions} \right) \quad (14)$$

Phase 1 is complete when $\text{twist} = 0$, $\text{flip} = 0$, and $\text{slice} = 0$.

3.2.2 Phase 2: $G_1 \rightarrow I$

In Phase 2, we solve from the subgroup state to the identity using only moves that preserve membership in H :

$$\text{Find } m_2 \in H^* \text{ such that } (s \cdot m_1) \cdot m_2 = I \quad (15)$$

The restricted move set $\{U, U^2, U', D, D^2, D', R^2, L^2, F^2, B^2\}$ ensures we remain within H .

The complete solution is the concatenation: $m = m_1 \cdot m_2$.

3.3 Coordinate System

The implementation uses a coordinate-based representation for efficient computation:

Coordinate	Range	Description
twist	0–2186	Corner orientation (3^7)
flip	0–2047	Edge orientation (2^{11})
slice	0–494	Middle edge positions ($\binom{12}{4}$)
URFtoDLF	0–20159	Corner permutation
FRtoBR	0–11879	Edge permutation subset
URtoDF	0–20159	Upper layer edge positions
parity	0–1	Permutation parity

Table 1: Coordinate representation for cube states

4 Methodology & Implementation

4.1 System Architecture

The solver system consists of three main components:

1. **Input Module:** Handles cube state acquisition via camera, manual entry, or 3D interface
2. **Solving Engine:** Implements Kociemba’s Two-Phase Algorithm
3. **Output Module:** Displays solutions with augmented reality visualization

4.1.1 Cube Representation

The implementation uses three complementary representations:

FaceCube (Facelet Level) A 54-character string representing each facelet:

```

1 # Solved cube representation
2 "UUUUUUUUURRRRRRRRRRFFFFFFFDDDDDDDDLLLLLLLLLLBBBBBBBB"
3
4 # Facelet positions:
5 #       U1 U2 U3
6 #       U4 U5 U6
7 #       U7 U8 U9
8 # L1 L2 L3 F1 F2 F3 R1 R2 R3 B1 B2 B3
9 # L4 L5 L6 F4 F5 F6 R4 R5 R6 B4 B5 B6
10 # L7 L8 L9 F7 F8 F9 R7 R8 R9 B7 B8 B9
11 #       D1 D2 D3
12 #       D4 D5 D6
13 #       D7 D8 D9

```

CubieCube (Cubie Level) Tracks 8 corners and 12 edges with position and orientation:

```

1 # Corner positions: URF, UFL, ULB, UBR, DFR, DLF, DBL, DRB
2 # Edge positions: UR, UF, UL, UB, DR, DF, DL, DB, FR, FL, BL, BR
3
4 cp = [0, 1, 2, 3, 4, 5, 6, 7] # Corner permutation
5 co = [0, 0, 0, 0, 0, 0, 0, 0] # Corner orientation (0-2)
6 ep = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11] # Edge permutation
7 eo = [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] # Edge orientation
    (0-1)

```

CoordCube (Coordinate Level) High-level abstraction for search algorithm with precomputed move tables.

4.2 IDA* Search Algorithm

The solver employs Iterative Deepening A* (IDA*), combining the space efficiency of depth-first search with the optimality of A*:

4.3 Pruning Tables (Heuristics)

The heuristic function $h(node)$ is implemented through precomputed pruning tables that store the minimum number of moves required to reach the goal state from any coordinate configuration.

4.3.1 Phase 1 Pruning Tables

- **Slice_Flip_Prun:** Combines slice and flip coordinates
- **Slice_Twist_Prun:** Combines slice and twist coordinates

4.3.2 Phase 2 Pruning Tables

- **Slice_URFtoDLF_Parity_Prun:** Corner permutation with parity
- **Slice_URtoDF_Parity_Prun:** Edge permutation with parity

The heuristic value is the maximum of the applicable pruning table lookups:

$$h(s) = \max(\text{Prun}_1(s), \text{Prun}_2(s)) \quad (16)$$

These tables are computed once and stored as binary files (approximately 50MB total), enabling $O(1)$ heuristic evaluation during search.

4.4 Computer Vision Module

The input module uses OpenCV for real-time color detection:

Algorithm 1 IDA* Search for Rubik's Cube

```
1: procedure IDA*(start)
2:   threshold  $\leftarrow h(\textit{start})$ 
3:   while threshold  $\leq \textit{maxDepth}$  do
4:     result  $\leftarrow \text{SEARCH}(\textit{start}, 0, \textit{threshold})$ 
5:     if result = FOUND then
6:       return solution
7:     end if
8:     threshold  $\leftarrow \textit{result}$ 
9:   end while
10:  return FAILURE
11: end procedure
12: procedure SEARCH(node, g, threshold)
13:  f  $\leftarrow g + h(\textit{node})$ 
14:  if f > threshold then
15:    return f
16:  end if
17:  if isGoal(node) then
18:    return FOUND
19:  end if
20:  min  $\leftarrow \infty$ 
21:  for each move in allowedMoves do
22:    child  $\leftarrow \textit{apply}(\textit{node}, \textit{move})$ 
23:    result  $\leftarrow \text{SEARCH}(\textit{child}, g + 1, \textit{threshold})$ 
24:    if result = FOUND then
25:      return FOUND
26:    end if
27:    min  $\leftarrow \min(\textit{min}, \textit{result})$ 
28:  end for
29:  return min
30: end procedure
```

```

1 # HSV-based color detection
2 def detect_color(h, s, v):
3     if s < 50:
4         return WHITE
5     elif h < 8:
6         return RED
7     elif h < 20:
8         return ORANGE
9     elif h < 40:
10        return YELLOW
11    elif h < 75:
12        return GREEN
13    elif h < 132:
14        return BLUE
15    else:
16        return RED # Red wraps around in HSV

```

The system samples 9 points per face (3×3 grid) and requires sequential scanning of all 6 faces using keyboard triggers (W, A, S, D, F, Z).

4.5 Technology Stack

Component	Technology	Purpose
Core Algorithm	Python 3 / C (CFFI)	Two-Phase solver
Computer Vision	OpenCV (cv2)	Camera capture, color detection
3D Visualization	Matplotlib	Interactive cube preview
AR Overlay	OpenCV	Move indication arrows
Numerics	NumPy	Array operations

Table 2: Technology stack

5 Results and Performance Analysis

5.1 Performance Metrics

The solver was evaluated on a test suite of 300+ scrambled cube configurations. Table 3 summarizes the key performance metrics.

5.2 Solution Quality Distribution

Analysis of solution lengths across the test suite reveals:

Metric	Two-Phase Solver	Layer-by-Layer
Average Move Count	19.2 moves	95–110 moves
Minimum Move Count	15 moves	80 moves
Maximum Move Count	24 moves	120 moves
Average Solve Time	48 ms	N/A (manual)
Maximum Solve Time	2.1 s	N/A
Memory Usage (Pruning Tables)	~50 MB	0 MB
Success Rate	100%	100%

Table 3: Performance comparison between Two-Phase Algorithm and Layer-by-Layer method

Move Count	Percentage of Solutions
15–17	12%
18–19	35%
20–21	38%
22–24	15%

Table 4: Distribution of solution lengths

5.3 Example Solutions

5.4 Comparison with God’s Number

The theoretical optimal (God’s Number = 20) guarantees that any cube can be solved in at most 20 moves. Our implementation achieves:

- **85%** of solutions within 21 moves
- **50%** of solutions at or below God’s Number (20 moves)
- **Average overhead:** 1.2 moves above optimal

This represents an excellent trade-off between solution quality and computational efficiency.

5.5 Real-Time Performance

For the computer vision pipeline:

6 Conclusion

This paper presented the implementation and analysis of a Rubik’s Cube solver combining Kociemba’s Two-Phase Algorithm with real-time computer vision capabilities. The key findings are:

1. **Efficiency:** The Two-Phase Algorithm consistently produces solutions averaging 19 moves in under 50 milliseconds, representing a $5\times$ improvement over Thistlethwaite’s algorithm and approaching the theoretical minimum of 20 moves.

Scramble State	Solution	Moves
BBURUDBFUFFRRFUUFL...	B U' L' D' R' D' L2 D' L F' L' D F2 R2 U R2 B2 U2 L2 F2 D'	20
DRLUUBFBRBLURRLRUBL...	D2 R' D' F2 B D R2 D2 R' F2 D' F2 U' B2 L2 U2 D R2 U	19
FLBUULFFFLFDURRDURBD...	U2 F' L F U' B2 U' D R2 F2 D2 L2 F2 D B2 D L2 D'	18

Table 5: Example solutions from test suite

Operation	Time
Frame capture	33 ms (30 FPS)
Color detection (per face)	5 ms
Cube validation	2 ms
Solution computation	48 ms (avg)
AR overlay rendering	8 ms
Total pipeline	<100 ms

Table 6: Real-time performance breakdown

2. **Practicality:** By precomputing pruning tables (50MB), the solver achieves $O(1)$ heuristic evaluation, enabling real-time performance suitable for interactive applications.
3. **Versatility:** The multi-modal input system (camera, manual, 3D) accommodates various use cases, from casual solving assistance to automated systems.
4. **Visualization:** The augmented reality overlay provides intuitive step-by-step guidance, making the solution accessible to users unfamiliar with cube notation.

While the Two-Phase Algorithm does not guarantee the absolute optimal solution (God’s Number = 20) for every configuration without extended computation, it provides near-optimal solutions almost instantaneously. This trade-off makes it the algorithm of choice for practical Rubik’s Cube solving applications.

The group-theoretic foundation—decomposing the solution into coset representatives followed by subgroup solving—represents an elegant application of abstract algebra to a concrete computational problem.

7 Future Scope

Several avenues exist for extending this work:

7.1 Enhanced AR Integration

- Real-time cube tracking with automatic face detection (eliminating manual scanning)
- 3D overlay of virtual cube aligned with physical cube

- Mobile application deployment using ARCore/ARKit

7.2 Robotic Automation

- Integration with stepper motor controllers for automated solving
- Optimization of move sequences for mechanical constraints (e.g., minimizing direction changes)
- Development of a complete robot solver system

7.3 Algorithm Extensions

- Adaptation to larger cubes (4×4 , 5×5) using reduction methods
- Implementation of optimal solving mode for configurations requiring extended computation
- Parallel search using GPU acceleration

7.4 Machine Learning Approaches

- Neural network-based heuristics for improved pruning
- Reinforcement learning for discovering novel solving strategies
- Deep learning for robust color detection under varying lighting conditions

References

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A Move Notation

Standard Rubik's Cube notation:

- **Single letter** (R, U, F, D, L, B): 90° clockwise rotation
- **Letter with apostrophe** (R', U', etc.): 90° counterclockwise rotation
- **Letter with 2** (R2, U2, etc.): 180° rotation

Face definitions:

- U (Up): White center
- D (Down): Yellow center
- F (Front): Blue center
- B (Back): Green center
- R (Right): Red center
- L (Left): Orange center

B Error Codes

The solver implements comprehensive validation:

- **Error 1:** Invalid color counts (each color must appear exactly 9 times)
- **Error 2:** Invalid edge configuration
- **Error 3:** Edge flip parity error
- **Error 4:** Invalid corner configuration
- **Error 5:** Corner twist parity error
- **Error 6:** Overall parity mismatch (unsolvable configuration)
- **Error 7:** Solution exceeds maximum depth
- **Error 8:** Timeout exceeded